

ALBERT ESSILFIE

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RESEARCH INTERESTS

1. Assistive Technology for Persons With Disabilities (PWDs)
2. Machine Learning in Robotics
3. Human Computer Interaction
4. Internet of Things in Agriculture

EDUCATION

University of Mines and Technology (UMaT), Ghana

BSc. Computer Science and Engineering, **First Class Honors (83% out of 100%)**

Relevant courses: Data Structures and Algorithm, Software Engineering, Embedded Systems, Report Writing, Database Management System, Operating System, Information Security, Simulation and Modelling, Microprocessors, Digital Hardware Design, Human Computer Interaction (HCI).

PROJECTS

Smart Navigation Device for the visually impaired(Capstone Project) May 2024 – Sept 2024

- Developed an app for visually impaired users, featuring object identification and caregiver tracking. Accra, Ghana
- Trained a model using real-world datasets for accurate object recognition.

Farmlynco Mobile App to link local farmers to consumers directly July 2024 - Oct 2024

- Created an online marketplace connecting farmers with suppliers and buyers. Addis, Ethiopia
- Led the integration of IoT weather station for providing farmers with accurate weather forecasts.
- Developed AI-driven "Crop Doctor" for diagnosing and managing crop diseases.

IoT-Enabled Multipurpose Emergency Detection System for Fire and Gas Leaks with Real-Time Alerts. Feb 2022 – Apr 2022 Tarkwa, Ghana

- Integrated SMS and phone call alerts for immediate notifications.
- Designed a user-friendly mobile app with voice command functionality for accessibility.

WORK EXPERIENCE

University of Ghana - Research Assistantship Oct 2024 - Present Accra Ghana

- Conducted research on assistive technologies aimed at improving accessibility for individuals with disabilities.
- Supported the development and implementation of prototypes and user-testing Procedures to evaluate the effectiveness of assistive devices.

University of Mines and Technology - Tarkwa, Western Networking Intern Oct - Dec 2022 Tarkwa, Ghana

- Gained hands-on experience in network administration and maintenance.
- Assisted in troubleshooting and resolving network issues, contributing to improved network reliability.

Inclusive Tech Group (ITG) – Graphics Designer Intern Oct - Dec 2021 Accra, Ghana

- Develop and maintain the visual identity and branding of ITG, which includes creating logos, color schemes, and design guidelines.
- Design graphics for promotional materials, social media posts, website banners and flyers.

SKILLS

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- **Graphic Designing:** Adobe PhotoShop
 - **UI / UX Designing:** Figma
 - **Flutter Mobile Application Developer**
 - **Programming languages:** JAVA, C++, Dart, Python.
 - **Engineering Tools and Softwares:** Arduino, Raspberry Pi, Visual Studio, Latex, ENSP Network Simulator, Wireshark, Proteus, Fritzing.

HONORS AND AWARDS

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| • Participated in the Africa to Silicon Valley AI for Impact Hackathon | Aug 2024 |
| • Participated in IndabaX Data Science Summit at KNUST | Jul 2024 |
| • Participated in Google Global Challenge | Jan 2023 |
| • Participated at the Tech in Ghana Conference by representing UMaT | May 2022 |

CERTIFICATIONS

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| • Huawei IoT | Jan 2024 |
| • Huawei Certified Network Associate – Routing and Switching. | Aug 2023 |
| • First runner-up Delta Innovation Camp | May 2023 |
| • University of Mines Innovation Fair, Winner | Jul 2022 |
| • First runner-up Disability Inclusive Hackathon | Dec 2021 |

VOLUNTEERING EXPERIENCE AND CO-CURRICULAR ACTIVITIES

Firm Health Ghana Foundation **September,2024**

- Served as a graphic designer and designed all the flyers for the blood donation exercise.
- Collected personal data of the people during the blood donation exercise.

Agri-IoT **August,2023**

- Contributed to the Agri-IoT project by implementing Arduino concepts, promoting innovation in agricultural technology as a volunteer.

Festive Kids Boot Camp **September,2023**

- Trained 300 pupils from the Cape Coast Municipality in robotics and programming, fostering STEM education and digital literacy within the community.

A Football Goalkeeper in the community team

LEADERSHIP EXPERIENCE

President St. Mary's Old Boys Association - UMaT **Sept 2023 - 2024**

- President, SMOBA - UMaT, showcasing strong leadership, training, and mentorship.

Financial Secretary, Aaenics UMaT Robotics Club **Jan 2023 - 2024**

- Created and managed the club's budget, ensuring expenses align with financial goals.

Organizing Secretary, Association of Computer Science and Engineering Student. **Oct 2022 - 2023**

- Planned and coordinated club events, meetings, and activities, ensuring they run smoothly.

PUBLICATIONS

Mohammed Y. Umaru, Emmanuel Effah, Minta Frederick, **Essilfie Albert**, "In-depth IoT-based Home Automation System for Ghana's Context", *International Journal of Research and Scientific Innovation (IJRSI)*. 2023

Abdul-Majeed Osman, **Albert Essilfie**, George Essah Yaw Okai, Minta Frederick Kwame, 'IoT-based Weather Monitoring System for Ghanaian Farmers', *International Journal of Computer Application (IJCA)*. 2024